

Smart Metering Portal System Administrator Manual

Infrastructure · Deployment · Maintenance

Version

1.0

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Audience

System Administrators / DevOps

Confidentiality

Internal Use Only

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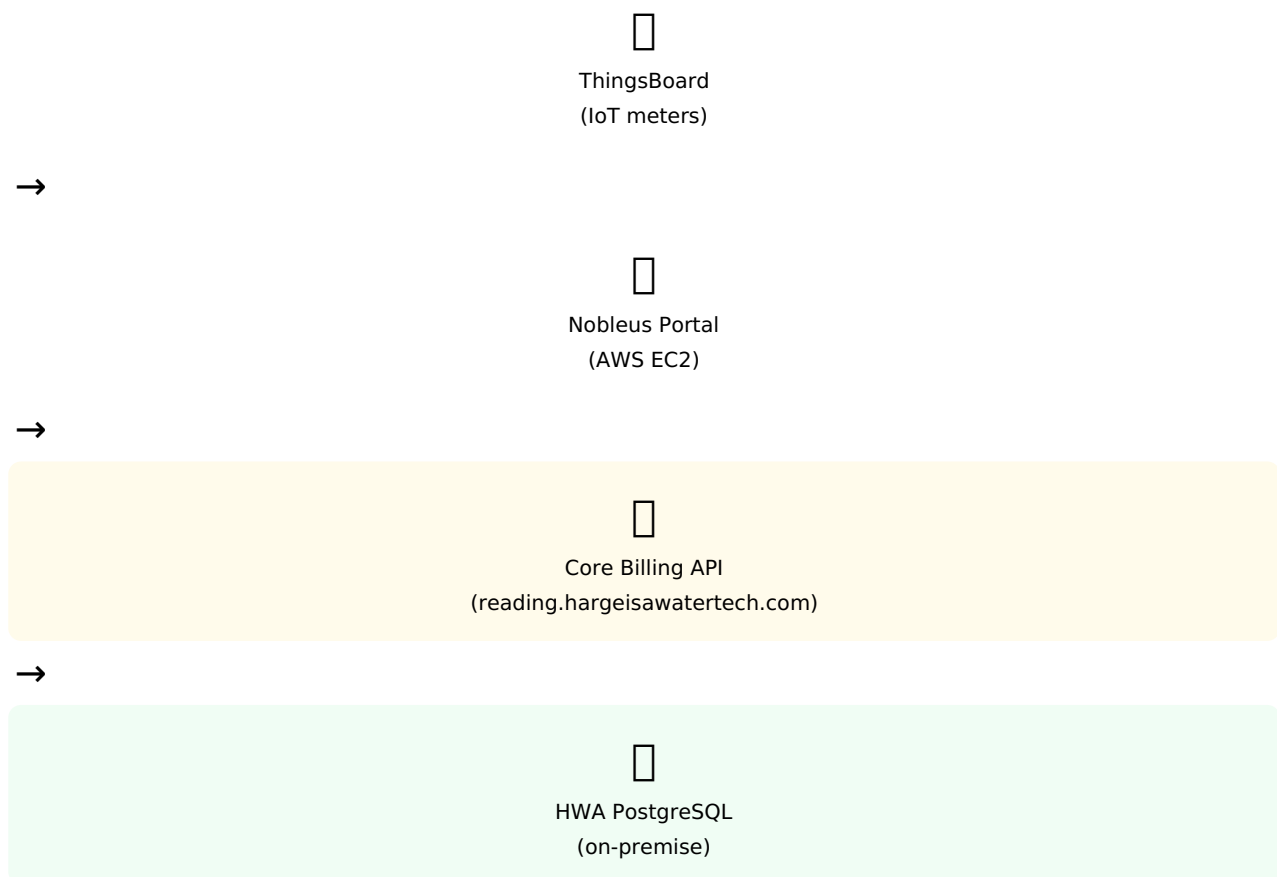
1. System Overview

The HWA Smart Metering Portal is a Yii 2 PHP web application hosted on AWS. It receives IoT water-meter readings via a ThingsBoard webhook, allows billing clerks to review and approve readings, and pushes approved readings to the core billing server

(reading.hargeisawatertech.com) for invoice generation.

portal.hargeisawatertech.com	Full admin/clerk portal (all modules)
billing.hargeisawatertech.com	Billing module only (restricted access)
api.hargeisawatertech.com	REST API for ThingsBoard & mobile apps
reading.hargeisawatertech.com	External HWA Core Billing API (remote)

2. Architecture



- ThingsBoard pushes telemetry via REST webhook to api.hargeisawatertech.com .
- The portal stores raw readings in its local MySQL database (nobleus).
- Billing clerks review and approve readings on the portal.
- Approved readings are pushed to reading.hargeisawatertech.com/bill/api/meter-reading .
- The core billing server writes invoices to the on-premise PostgreSQL database.

3. Server Requirements

OS	Ubuntu 22.04 LTS (Jammy) or later
Web Server	Apache 2.4 with <code>mod_rewrite</code> , <code>mod_ssl</code>
PHP	8.1 or 8.3 with extensions: <code>pdo_mysql</code> , <code>curl</code> , <code>mbstring</code> , <code>intl</code> , <code>gd</code> , <code>zip</code>
Database	MySQL 8.0 (local), PostgreSQL 10+ (remote — core billing)
Composer	2.x
Git	2.x
SSL	Let's Encrypt via Certbot

4. Environment Variables (.env)

All secrets and environment-specific values live in `/home/admin/nobleus-admin/.env` . This file is excluded from Git.

```
# — Database (local MySQL) —————
DB_HOST=localhost
DB_PORT=3306
DB_NAME=nobleus
DB_USER=nobleus
DB_PASSWORD=your_secure_password

# — ThingsBoard —————
TB_WEBHOOK_TOKEN=865e23dd298796... # webhook shared secret
TB_URL=https://hwa-smartmetering-app.com
TB_USERNAME=tenant@hargeisawatertech.com
TB_PASSWORD=tenant
TB_TELEMETRY_KEY=absoluteMeterReading_liters

# — Canonical Domains —————
BILLING_PORTAL_URL=https://billing.hargeisawatertech.com
PORTAL_URL=https://portal.hargeisawatertech.com
API_BASE_URL=https://api.hargeisawatertech.com

# — Core Billing API (remote) —————
CORE_READING_API_URL=https://reading.hargeisawatertech.com
CORE_METERS_API_URL=https://meters.hargeisawatertech.com
CORE_API_TOKEN=eyJ0eXAiOiJKV1Q... # JWT service token

# — Optional: disable SSL verify during cert issues —
# CORE_API_INSECURE_SSL=1
```

Warning

Never commit `.env` to Git. Rotate `CORE_API_TOKEN` and `TB_WEBHOOK_TOKEN` immediately if exposed.

5. Apache Virtual Hosts

Config files live in `/etc/apache2/sites-available/` . Enable with `a2ensite` .

<code>portal.hargeisawatertech.com-le-ssl.conf</code>	portal domain	backend/web
<code>nobleus-ssl.conf</code>	billing domain	backend/web
<code>api.hargeisawatertech.com.conf</code>	API domain	api/web
<code>000-catch-all.conf</code>	bare IP / unknown host	returns 403

Reload Apache after config changes

```
sudo systemctl reload apache2
```

Renew SSL certificates

```
sudo certbot renew --dry-run    # test first
sudo certbot renew              # apply
sudo systemctl reload apache2
```

6. Database

6.1 Local MySQL (nobleus)

<code>meter_reading_raw</code>	Raw IoT readings (liters) from ThingsBoard
<code>billing_ledger</code>	Clerk-approved readings with push status
<code>billing_audit_logs</code>	Audit trail for all import/push actions
<code>meters</code>	Physical meter registry
<code>meter_assignments</code>	Links meters to supply numbers

Run migrations

```
cd /home/admin/nobleus-admin
php yii migrate
```

Check push queue

```
mysql -u nobleus -p -D nobleus -e \  
"SELECT id, supply_no, push_status, pushed_at  
FROM billing_ledger  
WHERE push_status = 'failed'  
ORDER BY id DESC LIMIT 20;"
```

6.2 Remote PostgreSQL (HWA Core Billing)

- Host: 102.218.58.66 port 5432 (FortiGate VIP → 10.10.50.120)
- Database: HWA_DATA | User: openpg
- If the connection fails, readings queue locally and retry when the remote DB is restored.

Verify remote DB reachability

```
nc -vz 102.218.58.66 5432      # TCP test  
psql -h 102.218.58.66 -U openpg -d HWA_DATA -c "SELECT version();"
```

7. Billing Date Policy (28th Rule)

Company Policy

All meter readings posted to the core billing server use

the 28th of the reading's month

as the official billing date, regardless of when the reading was physically taken.

This policy is enforced in code in two places:

- **Bulk post** (`actionPostToBilling`) — uses `$month . '-28'` as the target date.
- **Single post** (`actionImport`) — derives `date('Y-m', ...) . '-28'` from the reading's timestamp.

If the reading was taken on a different day (e.g., the 7th), a `description` field is automatically appended to the API payload noting the actual reading date. This note appears in the core billing system for auditors.

8. Deployment & Upgrade

8.1 First-time setup

Clone the repository: `git clone git@github.com:apitab/nobleus-admin.git /home/admin/nobleus-admin`

Copy and edit the environment file: `cp .env.example .env && nano .env`

Install PHP dependencies: `composer install --no-dev --optimize-autoloader`

Run database migrations: `php yii migrate`

Set web directory permissions: `chmod -R 775 backend/web/assets backend/runtime common/runtime`

Enable Apache vhosts and reload: a2ensite portal.hargeisawatertech.com-le-ssl.conf nobleus-ssl.conf
api.hargeisawatertech.com.conf && sudo systemctl reload apache2

8.2 Code upgrade

```
cd /home/admin/nobleus-admin
git pull origin main
composer install --no-dev --optimize-autoloader
php yii migrate          # apply any new migrations
sudo systemctl reload apache2 # reload Apache (view changes take effect immediately)
```

Note

PHP view files take effect on the next page load without a restart. Only Apache config changes require reload .

9. Logs & Monitoring

Apache access	/var/log/apache2/portal-access.log
Apache error	/var/log/apache2/portal-error.log
Application (Yii)	/home/admin/nobleus-admin/backend/runtime/logs/app.log
Catch-all requests	/var/log/apache2/catchall-error.log

Tail application errors in real time

```
tail -f /home/admin/nobleus-admin/backend/runtime/logs/app.log
```

Check failed billing pushes in audit log

```
mysql -u nobleus -p -D nobleus -e \
"SELECT id, action, supply_no, details, created_at
FROM billing_audit_logs
WHERE action LIKE '%failed%'
ORDER BY id DESC LIMIT 20;"
```

10. Troubleshooting Reference

Push status stays failed	Remote billing DB unreachable	Check <code>nc -vz 102.218.58.66 5432</code> ; contact remote DB admin; run <code>retry</code> command once restored
Apache shows SSL cert error on reload	Let's Encrypt cert missing or expired	<code>sudo certbot renew && sudo systemctl reload apache2</code>
504 Gateway Timeout on bulk post	Remote API too slow / many meters	Increase <code>CURLOPT_TIMEOUT</code> in <code>CoreBillingClient.php</code> ; run in smaller month batches
Readings not appearing in portal	ThingsBoard webhook failing	Check <code>TB_WEBHOOK_TOKEN</code> in <code>.env</code> ; verify ThingsBoard rule-chain target URL
Zero readings in ledger	Meter transmitting 0 liters	Check meter on-site; zero readings are intentionally skipped and never posted to billing
<code>billing_audit_logs</code> empty	Table missing (old install)	Run <code>php yii migrate</code> to create the table

11. Security Checklist

- Rotate `CORE_API_TOKEN` every 6 months or after any staff departure.
- Rotate `TB_WEBHOOK_TOKEN` every 6 months.
- Keep `.env` permissions to `640` (owner read/write, group read only).
- Ensure `YII_DEBUG=false` and `YII_ENV=prod` in production `index.php` .
- SSL certificates must not expire — set up auto-renewal cron: `0 3 * * * certbot renew --quiet` .
- Restrict MySQL user `nobleus` to `localhost` only.
- The catch-all Apache vhost blocks all bare-IP access — do not disable it.
- Enable AWS Security Group rules: allow 80, 443 inbound; restrict 22 (SSH) to trusted IPs.